$$a_3 = \frac{n-1}{n+1} a_2 = \frac{0.36}{2.36} \times 0.874a \approx 0.129a$$

$$a_5 = \frac{2n}{n+1} a_3 = \frac{2 \times 1.36}{2.36} \times 0.129a \approx 0.149a$$

$$a_4 = \frac{2n}{1+n} a_2 = \frac{2 \times 1.36}{2.36} \times 0.847a \approx 0.977a$$

$$a_7 = \frac{2n}{n+1} a_6 = \frac{2n}{n+1} \cdot \frac{n-1}{n+1} a_3 = \frac{2 \times 1.36 \times .36}{(2.36)^2} a_3$$

$$\approx 0.023a$$

We first note that the sign of a_5 is opposite to that of a_1 which is a consequence of the fact that a sudden phase change of π occurs when the ray gets reflected at the point B. Further the magnitude of a_5 is nearly equal to that of a_1. On the other hand $|a_7| << |a_4|$. This is the reason why the interference fringes formed in transmission have poor contrast.

15.9 COLORS OF THIN FILMS

LO 7

We have seen in the previous section that if light from an extended monochromatic source (like a sodium lamp) is incident normally on a wedge, then equally spaced dark and bright fringes will be observed. The distance between two consecutive bright (or dark) fringes is determined by the wedge angle, the wavelength of light and by the refractive index of the film. If we use a polychromatic source (like an incandescent lamp) we will observe colored fringes. Further, if instead of a wedge we have a film of arbitrarily varying thickness we will again observe fringes, each fringe representing the locus of constant film thickness (see Fig. 15.28). This is indeed what we see when sunlight falls on a soap bubble or on a thin film of oil on water; see Fig. 13 in the prelim pages. It should be mentioned that if the optical path difference between the waves reflected from the upper surface of the film and from the lower surface of the film exceeds a few wavelengths, the interference pattern will be washed out due to the overlapping of interference patterns of many colors and no fringes will be seen (see Sec. 14.9). Thus, in order to see the fringes with white light, the film should not be more than few wavelength thick.



Fig. 15.28 A typical fringe pattern produced by an airfilm formed between two glass surfaces (which are not optically flat) and placed in contact with each other. Whenever the thickness of the airfilm is $m\lambda/2$, we obtain a dark fringe and when the thickness is $\left(m+\frac{1}{2}\right)\lambda/2$, we obtain a bright fringe. Each fringe describes a focus of equal thickness of the film. [*Photograph courtesy: Prof. R.S. Sirohi*].

15.10 NEWTON'S RINGS

If we place a plano-convex lens on a plane glass surface, a thin film of air is formed between the curved surface of the lens (AOB) and the plane glass plate (POQ)—see Fig. 15.29.

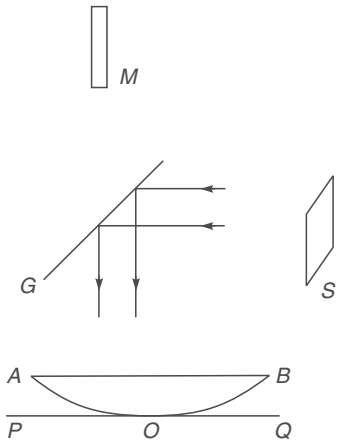


Fig. 15.29 An arrangement for observing Newton's rings. Light from an extended source S is allowed to fall on a thin film formed between the plano-convex lens AOB and the plane glass plate POQ. M represents a traveling microscope.

The thickness of the air film is zero at the point of contact O and increases as one moves away from the point of contact. If we allow monochromatic light (such as from a sodium lamp) to fall on the surface of the lens, then the light reflected from the surface AOB interferes with the light reflected from the surface POQ. For near normal incidence (and considering points very close to the point of contact) the optical path difference between the two waves is very nearly equal to $2nt$, where n is the refractive index of the film and t the thickness of the film. Thus, whenever the thickness of the air film satisfies the condition

$$2nt = \left(m + \frac{1}{2}\right)\lambda;\ m = 0, 1, 2,\ldots \qquad \textbf{(15.62)}$$

we will have maxima. Similarly, the condition

$$2nt = m\lambda \qquad \textbf{(15.63)}$$

will correspond to minima. Since the convex side of the lens is a spherical surface, the thickness of the air film will be constant over a circle (whose center will be at O) and we will obtain concentric dark and bright rings. These rings are known as **Newton's rings**.* It should be pointed out that in order to observe the fringes, the microscope (or the eye) has to be focussed on the upper surface of the film (see the discussion in Sec. 15.7).

The radii of various rings can easily be calculated. As mentioned earlier, the thickness of the air film will be constant over a circle whose center is at the point of contact O. Let the radius of the mth dark ring be r_m and if t is the thickness of the air film where the mth dark ring appears to be formed, then

$$r_m^2 = t(2R - t) \qquad \textbf{(15.64)}$$

where R represents the radius of curvature of the convex surface of the lens (see Fig. 15.30). Now $R \approx 100$ cm and $t \lesssim 10^{-3}$ cm, thus we may neglect t in comparison to $2R$ to obtain

$$r_m^2 \approx 2Rt$$

or

$$2t \approx \frac{r_m^2}{R} \qquad \textbf{(15.65)}$$

Substituting this in Eq. (15.63), we get

$$r_m^2 \approx m\lambda R\ ;\ m = 0, 1, 2,\ldots \qquad \textbf{(15.66)}$$

which implies that the radii of the rings vary as square root of natural numbers. Thus the rings will become close to each other as the radius increases (see Fig. 15.31). Between the two dark rings there will be a bright ring whose radius will be $\sqrt{m + \frac{1}{2}}\ \lambda R$.

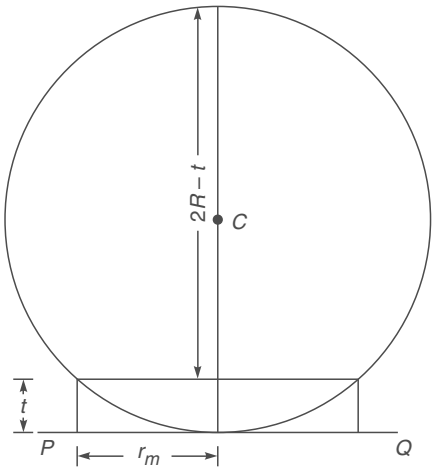


Fig. 15.30 r_m represents the radius of the mth dark ring; the thickness of the air film (where the mth dark ring is formed) is t.

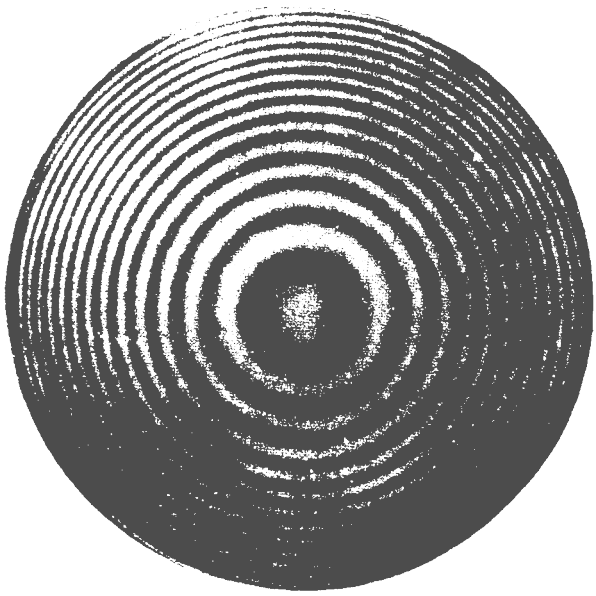


Fig. 15.31 Newton's rings as observed in reflection. The rings observed with transmitted light are of much poorer contrast. [*Photograph courtesy: Dr. G. Bose*].

Newton's rings can easily be observed in the laboratory by using an apparatus as shown in Fig. 15.29. Light from an extended source (emitting almost monochromatic light, like a sodium lamp) is allowed to fall on a glass plate which partially reflects the beam. This reflected beam falls on the plano-convex lens–glass plate arrangement and Newton's rings can easily be observed by viewing directly or through

* Boyle and Hooke had independently observed the fringes earlier but Newton was the first to measure their radii and make an analysis. The proper explanation was given by Thomas Young. Also see 'Milestones' in the beginning of this chapter.

a travelling microscope M. Actually, one really need not have plano-convex lens; the rings would be visible even when a biconvex lens is used.

Typically for $\lambda = 6 \times 10^{-5}$ cm, and $R = 100$ cm

$$r_m = 0.0774 \sqrt{m} \text{ cm} \qquad \textbf{(15.67)}$$

Thus the radii of the first, second and third dark rings would be approximately 0.0774 cm, 0.110 cm and 0.134 cm, respectively. Notice that the spacing between the second and third dark rings is smaller than the spacing between the first and second dark rings.

Equation (15.63) predicts that the central spot should be dark. Normally, with the presence of minute dust particles the point of contact is really not perfect and the central spot may not be perfectly dark. Thus, while carrying out the experiment one should measure the radii of the mth and the $(m+p)$th ring ($p \approx 10$) and take the difference in the squares of the radii ($r^2_{m+p} - r^2_m = p\lambda R$), which is indeed independent of m. Usually, the diameter can be more accurately measured and in terms of the diameters the wavelength is given by the following expression:

$$\lambda = \frac{D^2_{m+p} - D^2_m}{4pR} \qquad \textbf{(15.68)}$$

The radius of curvature can be accurately measured with the help of a spherometer and therefore by carefully measuring the diameters of dark (or bright) rings one can experimentally determine the wavelength.

It may be mentioned that if a liquid of refractive index n is introduced between the lens and the glass plate, the radii of the dark rings would be given by

$$r_m = (m\lambda R/n)^{1/2} \qquad \textbf{(15.69)}$$

Equation (15.69) may be compared with Eq. (15.66). Further, if the refractive indices of the material of the lens and of the glass plate are different and if the refractive index of the liquid lies in between the two values, the central spot will be bright as in Fig. 15.31 and Eq. (15.69) would give the radii of the bright rings.

An important practical application of the principle involved in the Newton's rings experiment lies in the determination of the optical flatness of a glass plate. Consider a glass surface placed on another surface whose flatness is known. If a monochromatic light beam is allowed to fall on this combination and the reflected light is viewed by a microscope, then, in general, dark and bright patches will be seen (Fig. 15.28). The space between the two glass surfaces forms an air film of varying thickness and whenever this thickness becomes $m\lambda/2$, we see a dark spot and when this thickness becomes $\left(m + \frac{1}{2}\right) \lambda/2$ we see a bright spot. Two consecutive dark fringes will be separated by the air film whose thickness will differ by $\lambda/2$. Consequently, by measuring the distance between consecutive dark and bright fringes one can calculate the optical flatness of a glass plate.

When we observe Newton's rings using a white light source, we will have situation similar to that discussed in Sec. 14.9, i.e., we will see only few colored fringes. However, if we put a red filter in front of our eye, the fringe pattern (corresponding to the red color) will suddenly appear. If we replace the red filter by a green filter in front of our eye, the fringe pattern corresponding to the green color will appear; this is similar to the discussion we had in Sec. 14.9.

Example 15.2 Consider the formation of Newton's rings by monochromatic light of $\lambda = 6.4 \times 10^{-5}$ cm. Assume the point of contact to be perfect. Now slowly raise the lens vertically above the plate. As the lens moves gradually away from the plate, discuss the ring pattern as seen through the microscope. Assume the radius of the convex surface to be 100 cm.

Solution: Since the point of contact is perfect, the central spot will be dark, the first dark ring will form at P where $PA = \lambda/2$, and the radius of this ring, OA, will be $\sqrt{\lambda R}$ (= 0.080 cm) –see Fig. 15.32(a). Similarly, the radius of the second dark ring will be $OB = \sqrt{2\lambda R}$ (= 0.113 cm). If we now raise the lens by $\frac{\lambda}{4}$(=1.6×10^{-5} cm) then $2t$ corresponding to the central spot would be $\lambda/2$ and instead of the dark spot at the centre we will now have a bright spot. The radii of the first and the second dark rings will be

$$OA_1 = \left(\frac{1}{2}\lambda R\right)^{1/2} = 0.0566 \text{ cm}$$

and

$$OB_1 = \left(\frac{3}{2}\lambda R\right)^{1/2} = 0.098 \text{ cm}$$

respectively [see Fig. 15.32(b)]. If the lens is further moved by $\lambda/4$ (see Fig. 15.32(c)], then the first dark ring collapses to the center and the central spot will be dark. The ring which was originally at Q now shifts to Q_2; similarly the ring at R [Fig. 15.32(a)] collapses to R_2 [Fig. 15.32(c)].

Thus, as the lens is moved upward the rings collapse to the centre. Hence if we can measure the distance by which the lens is moved upward and also count the number of dark spots that have collapsed to the center, we can determine the wavelength. For example, in the present case, if the lens is moved by 6.4×10^{-3} cm, 200 rings will collapse to the center. If one carries out this experiment it will be observed that the 200th dark ring will slowly converge to the center and when the lens has moved exactly by 6.4×10^{-3} cm it has exactly come to the center.

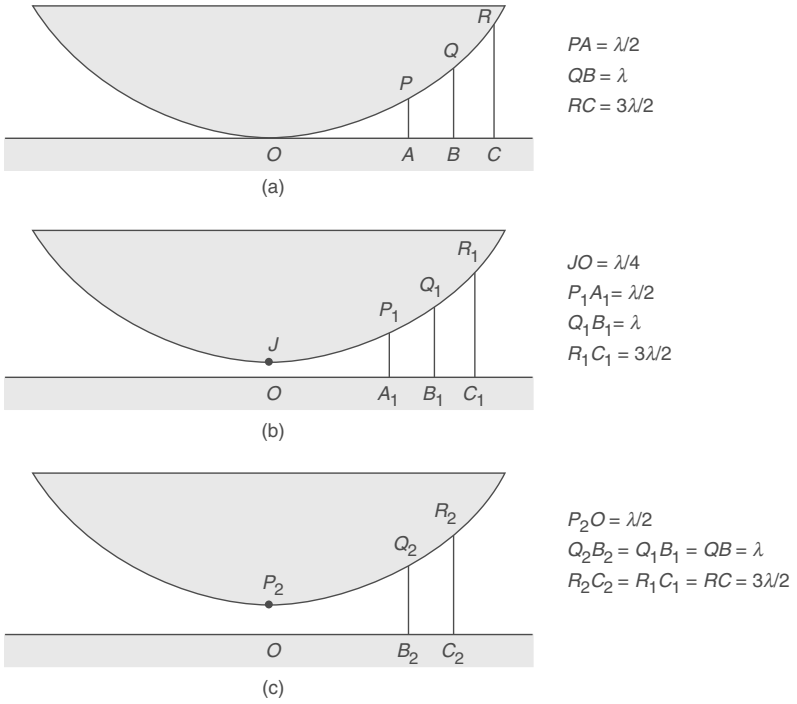


Fig. 15.32 The rings collapse to the center as the lens is moved away from the plate.

Example 15.3 Consider the formation of Newton's rings when two closely spaced wavelengths are present; for example, the D_1 and D_2 lines of sodium (λ_1 = 5890 Å and λ_2 = 5896 Å). What will be the effect of the presence of these two wavelengths as the lens is gradually moved away from the plate? What will happen if the sodium lamp is replaced by a white light source?

Solution: We will first assume that the lens is in contact with the plane glass plate [see Fig. 15.32(a)]. Since the two wavelengths are very close, the bright and dark rings of λ_1 superpose on the bright and dark rings of λ_2, respectively. This can easily be seen by calculating the radii of the ninth dark and bright ring for each wavelength.

For $\lambda = 5.890 \times 10^{-5}$ cm,

$$\text{radius of the ninth bright ring} = \sqrt{\left(9+\frac{1}{2}\right)\lambda R}$$

$$= \sqrt{9.5 \times 5.890 \times 10^{-5} \times 100}$$

$$= 0.236548 \text{ cm}$$

$$\text{radius of the ninth dark ring} = \sqrt{9\lambda R}$$

$$= 0.230239 \text{ cm}$$

Similarly, for $\lambda = 5.896 \times 10^{-5}$ cm,

$$\text{radius of the ninth bright ring} = \sqrt{9.5 \times 5.896 \times 10^{-5} \times 100}$$

$$= 0.236669 \text{ cm}$$

and

$$\text{radius of the ninth dark ring} = \sqrt{9 \times 5.896 \times 10^{-3}}$$

$$= 0.230356 \text{ cm}$$

Thus the rings almost exactly superpose on each other. However, for large values of m, the two ring patterns may produce uniform illumination. To be more specific, when the air film thickness t is such that

$$2t = m\lambda_1 = \left(m+\frac{1}{2}\right)\lambda_2$$

or
$$\frac{2t}{\lambda_2} - \frac{2t}{\lambda_1} = \frac{1}{2} \tag{15.70}$$

then around that point the fringe system will completely disappear; i.e., the bright ring for the wavelength λ_1 will fall on the dark ring for the wavelength λ_2 and conversely. Thus the contrast will be zero and no fringe pattern will be visible. Rewriting Eq. (15.70) we get

$$2t\,\frac{\lambda_1 - \lambda_2}{\lambda_1 \lambda_2} = \frac{1}{2}$$

or
$$2t = \frac{1}{2}\frac{\lambda_1\lambda_2}{\Delta\lambda} \approx \frac{1}{2}\frac{(5.893 \times 10^{-5})^2}{6 \times 10^{-8}}$$

$$\approx 3 \times 10^{-2} \text{ cm}$$

This will correspond to $m \approx 500$.

We shall see the effect of the same phenomenon if we slowly raise the convex lens in the upward direction as we had considered in Example 15.2. Let t_0 be the vertical distance through which the lens has been raised (see Fig. 15.33) and let t_0 be such that it satisfies the following equation:

$$\frac{2t_0}{\lambda_2} - \frac{2t_0}{\lambda_1} = \frac{1}{2}$$

or

$$t_0 = \frac{\lambda_1\lambda_2}{4(\lambda_1 - \lambda_2)}$$

Thus, if the point J (see Fig. 15.33) corresponds to a dark spot for λ_1 then it will correspond to a bright spot for λ_2 and conversely. Further, the nearby dark rings for λ_1, will almost fall at the same place as the bright rings for λ_2 and the interference pattern will be washed out. Thus viewing from a microscope we will not be able to see any ring pattern. Now, if the lens is further moved upwards by a distance t_0, then we will have

$$\frac{2t_1}{\lambda_2} - \frac{2t_1}{\lambda_1} = 1 \qquad \textbf{(15.71)}$$

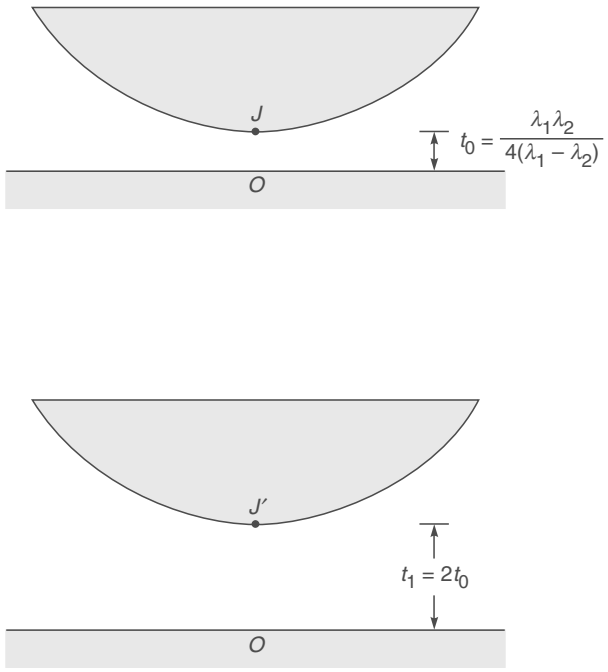


Fig. 15.33 In the Newton's rings experiment, if the light consists of two closely spaced wavelengths λ_1 and λ_2 (like the D_1 and D_2 lines of sodium) then, if the lens is separated by a distance t_0 $\left(= \frac{\lambda_1\lambda_2}{4(\lambda_1 - \lambda_2)}\right)$ interference fringes will be washed out. The fringes will reappear when the distance is $2t_0$.

where $t_1 = 2t_0$. Consequently, if the point J' corresponds to a dark spot for λ_1 then it will also correspond to a dark spot for λ_2. The fringe pattern will reappear but now with a slightly weaker contrast (see also Chapter 17).

In this way if we continue to move the lens upwards the fringe system will reappear every time the lens is moved up by a distance $2t_0$ $\left(\approx \frac{1}{2}\frac{\lambda_1\lambda_2}{\Delta\lambda}\right)$. This principle is used in a Michelson interferometer to measure the small wavelength difference $\Delta\lambda$, between two closely spaced lines (like the D_1 and D_2 lines of sodium).

It should be pointed out that for complete disappearance of the fringe pattern the intensities of the two lines λ_1 and λ_2 should be the same.

Another corollary of the above experiment consists in finding the change in the interference pattern (as we move up the convex lens) when we consider a single line of wavelength λ, but which has a width of $\Delta\lambda$. Thus we should assume all wavelengths between λ and $\lambda + \Delta\lambda$ to exist. By finding the approximate height at which the fringes disappear one can calculate $\Delta\lambda$. The coherence length (L) is related to $\Delta\lambda$ through the following relation (see Sec. 17.2):

$$L \sim \frac{\lambda^2}{\Delta\lambda} \qquad \textbf{(15.72)}$$

15.11 THE MICHELSON INTERFEROMETER* LO 10

A schematic diagram of the Michelson interferometer is shown in Fig. 15.34. S represents a light source (which may be a sodium lamp) and L represents a ground glass plate so that an extended source of almost uniform intensity is formed. G_1 is a beam splitter; i.e., a beam incident on G_1 gets partially reflected and partially transmitted. M_1 and M_2 are good quality plane mirrors having very high reflectivity. One of the mirrors (usually M_2) is fixed and the other (usually M_1) is capable of moving away or towards the glass plate G_1 along an accurately machined track by means of a screw. In the normal adjustment of the interferometer, the mirrors M_1 and M_2 are perpendicular to each other and G_1 is at 45° to the mirror.

Waves emanating from a point P get partially reflected and partially transmitted by the beam splitter G_1, and the two resulting beams are made to interfere in the following manner: The reflected wave [shown as (1) in Fig. 15.34] undergoes a further reflection at M_1 and this reflected wave gets (partially) transmitted through G_1; this is shown as (5) in the figure. The transmitted wave [shown as (2) in Fig. 15.34] gets reflected by M_2 and gets (partially) reflected by G_1 and results in the wave shown as (6) in the figure.

* The Michelson – Morley experiments using the Michelson interferometer to detect the motion of the earth with respect to the "ether" is discussed in Chapter 31.

Waves (5) and (6) interfere in a manner exactly similar to that shown in Fig. 15.22. This can easily be seen from the fact that if x_1 and x_2 are the distances of the mirrors M_1 and M_2 from the plate G_1, then to the eye the waves emanating from the point P will appear to get reflected by two parallel mirrors [M_1 and M_2' – see Fig. 15.34] separated by a distance $(x_1 \sim x_2)$. As discussed in Sec. 15.7, if we use an extended source, then no definite interference pattern will be obtained on a photographic plate placed at the position of the eye. Instead, if we have a camera focused for infinity, then on the focal plane we will obtain circular fringes, each circle corresponding to a definite value of θ (see Figs. 15.22 and 15.35); the circular fringes will look like the ones shown in Fig. 15.36. Now, if the beam splitter is just a simple glass plate, the beam reflected from the mirror M_2 will undergo an abrupt phase change of π (when getting reflected by the beam splitter) and since the extra path that one of the beams will traverse will be $2(x_1 \sim x_2)$, the condition for destructive interference will be

$$2\, d \cos\theta = m\, \lambda$$

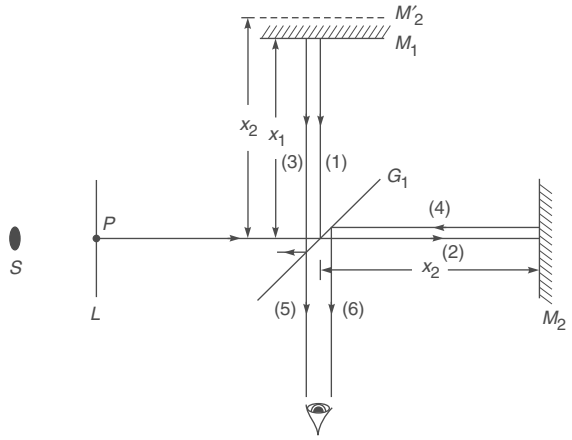


Fig. 15.34 Schematic of the Michelson interferometer.

where $m = 0, 1, 2, 3, \ldots\ldots$ and

$$d = x_1 \sim x_2$$

and the angle θ represents the angle that the rays make with the axis (which is normal to the mirrors as shown in Fig. 15.35). Similarly, the condition for a bright ring would be

$$2\, d \cos\theta = \left(m + \frac{1}{2}\right)\lambda$$

For example, for $\lambda = 6 \times 10^{-5}$ cm if $d = 0.3$ mm, the angles at which the dark rings will occur will be

$$\theta = \cos^{-1}\left(\frac{m}{1000}\right)$$

$$= 0^\circ, 2.56^\circ, 3.62^\circ, 4.44^\circ, 5.13^\circ, 5.73^\circ, 6.28^\circ, \ldots$$

corresponding to $m = 1000, 999, 998, 997, 996, 995, \ldots$ Thus the central dark ring in Fig. 15.36(a) corresponds to $m = 1000$, the first dark ring corresponds to $m = 999$, etc. If we now reduce the separation between the two mirrors so that $d = 0.15$ mm, the angles at which the dark rings will occur will be [see Fig.15.36(b)]

$$\theta = \cos^{-1}\left(\frac{m}{500}\right) = 0^\circ, 3.62^\circ, 5.13^\circ, 6.28^\circ, 7.25^\circ, \ldots$$

where the angles now correspond to $m = 500, 499, 498, 497, 496, 495, \ldots$Thus as we start reducing the value of d, the fringes will appear to collapse at the centre and the fringes become less closely placed. It may be noted that if d is now slightly decreased, say from 0.15 mm to 0.14985 mm,

$$2\, d = 499.5\, \lambda$$

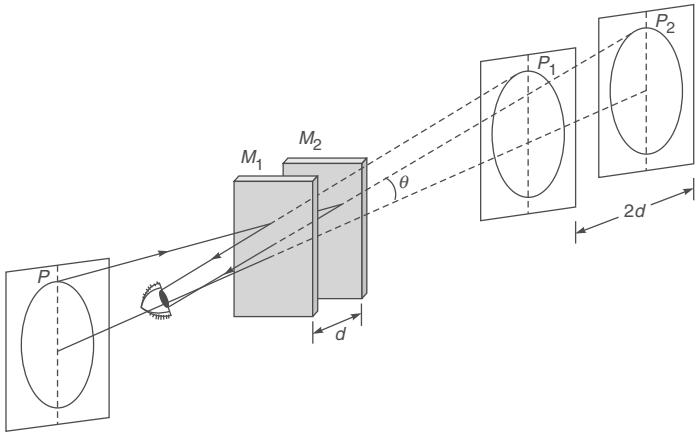


Fig. 15.35 A schematic of the formation of circular fringes [Adapted from Ref. 15.7].

the dark central spot in Fig.15.36(b) (corresponding to m = 500) would disappear and the central fringe will become bright. Thus, as d decreases, the fringe pattern tends to collapse towards the centre. (Conversely, if d is increased, the fringe pattern will expand.) Indeed, if N fringes collapse to the centre as the mirror M_1 moves by a distance d_0, then we must have

$$2\,d = m\,\lambda$$
$$2\,(d - d_0) = (m - N)\,\lambda$$

where we have put $\theta' = 0$ because we are looking at the central fringe. Thus,

$$\lambda = \frac{2d_0}{N} \qquad \textbf{(15.73)}$$

This provides us with a method for the measurement of the wavelength. For example, in a typical experiment, if one finds 1000 fringes collapse to the center as the mirror is moved through a distance of 2.90×10^{-2} cm, then

$$\lambda = 5800 \text{ Å}$$

The above method was used by Michelson for the standardization of the meter. He had found that the red cadmium line (λ = 6438.4696 Å) is one of the ideal monochromatic sources and as such this wavelength was used as a reference for the standardization of the meter. In fact, he defined the meter by the following relation:

1 meter = 1553164.13 red cadmium wavelengths,

the accuracy is almost one part in 10^9.

In an actual Michelson interferometer, the beam splitter G_1 consists of a plate (which may be about 1/2 cm thick), the back surface of which is partially silvered and the reflections occur at the back surface as shown in Fig. 15.37. It is immediately obvious that the beam (5) traverses the glass plate thrice and in order to compensate for this additional path, one introduces a 'compensating plate' G_2 which is exactly of the same thickness as G_1. The compensating plate is not really necessary for a monochromatic source because the additional path $2(n - 1)t$ introduced by G_1 can be compensated by moving the mirror M_1 by a distance $(n - 1)t$ where n is the refractive index of the material of the glass plate G_1.

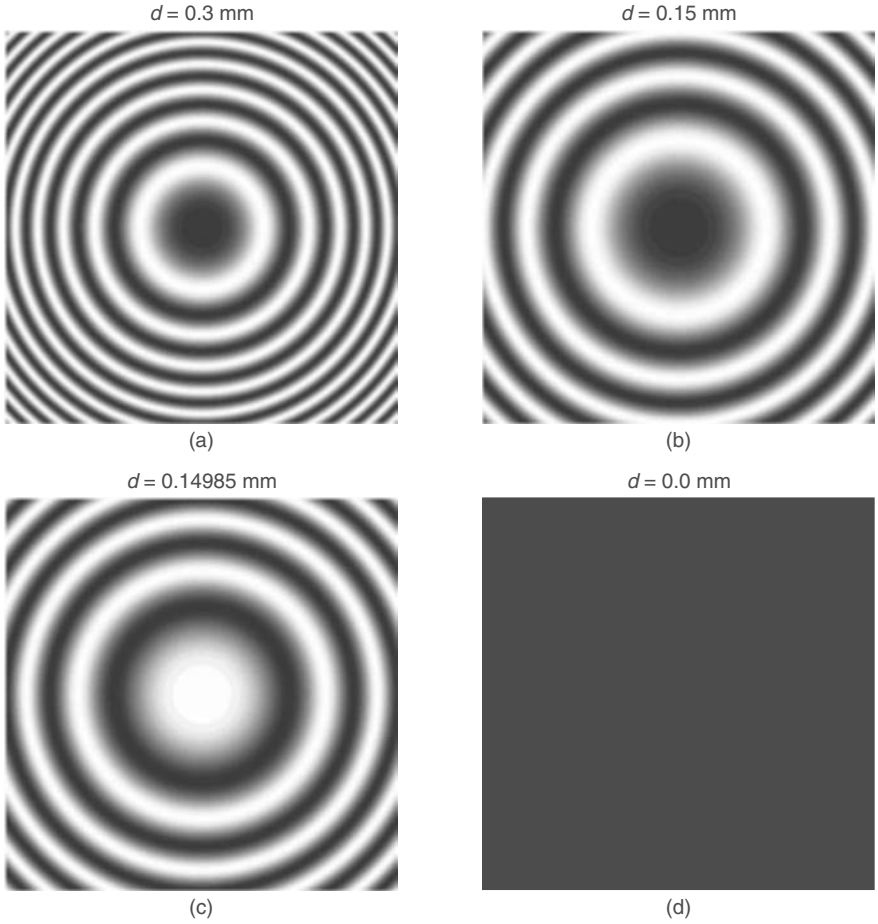


Fig. 15.36 Computer generated interference pattern produced by a Michelson interferometer.

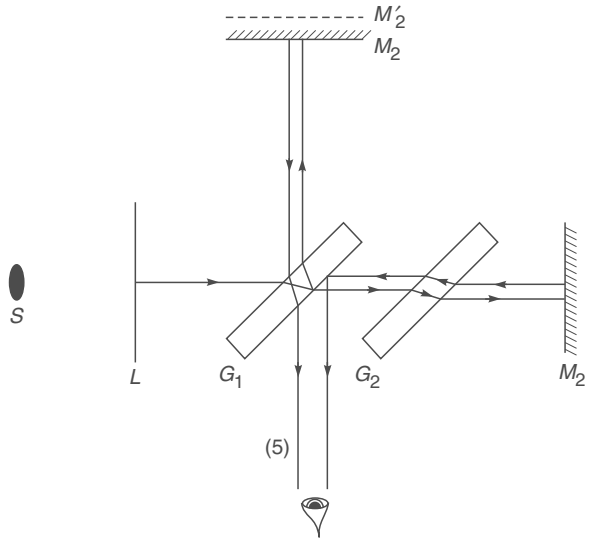


Fig. 15.37 In an actual interferometer there is also a compensating plate G_2.

However, for a white light source it is not possible to simultaneously satisfy the zero path-difference condition for all wavelengths, since the refractive index depends on wavelength. For example, for $\lambda = 6560$ Å and 4861 Å, the refractive index of crown glass is 1.5244 and 1.5330 respectively. If we are using a 0.5 cm thick crown glass plate as G_1, then M_1 should be moved by 0.2622 cm for λ = 6560 Å and by 0.2665 cm for λ = 4861 Å, the difference between the two positions corresponding to over hundred wavelengths! Thus, if we have a continuous range of wavelengths from 4861 Å to 6560 Å, the path difference between any pair of interfering rays (see Fig. 15.34) will vary so rapidly with wavelength that we would observe only a uniform white light illumination. However, in the presence of the compensating plate G_2, one would observe a few colored fringes around the point corresponding to zero path difference (see Sec. 14.9).

Michelson interferometer can also be used in the measurement of two closely spaced wavelengths. Let us assume that we have a sodium lamp which emits predominantly two closely spaced wavelengths 5890 Å and 5896 Å. The interferometer is first set corresponding to the zero path difference.* Near $d = 0$, both the fringe patterns will overlap. If the mirror M_1 is moved away (or towards) the plate G_1 through a distance d, then the maxima corresponding to the wavelength λ_1 will not, in general, occur at the same angle as λ_2. Indeed, if the distance d is such that

$$\frac{2d}{\lambda_1} - \frac{2d}{\lambda_2} = \frac{1}{2} \qquad \textbf{(15.74)}$$

and if $2d \cos \theta' = m\lambda_1$, then $2d \cos \theta' = \left(m + \frac{1}{2}\right)\lambda_2$. Thus, the maxima of λ_1 will fall on the minima of λ_2 and conversely, and the fringe system will disappear. It can easily be seen that if

$$\frac{2d}{\lambda_1} - \frac{2d}{\lambda_2} = 1 \qquad \textbf{(15.75)}$$

then interference pattern will again reappear. In general, if

$$\frac{2d}{\lambda_1} - \frac{2d}{\lambda_2}$$

is 1/2, 3/2, 5/2,…we will have disappearance of the fringe pattern and if it is equal to 1, 2, 3,…then the interference pattern will appear.

Instead of two discrete wavelengths, if the source consists of all wavelengths, lying between λ and $\lambda + \Delta\lambda$, then no interference pattern will be observed if

$$\frac{2d}{\lambda} - \frac{2d}{\lambda + \frac{\Delta\lambda}{2}} \gtrsim \frac{1}{2}$$

or

$$2d \gtrsim \frac{\lambda^2}{\Delta\lambda} \qquad \textbf{(15.76)}$$

In this case the fringes will not reappear because we have a continuous range of wavelengths rather than two discrete wavelengths (see Sec. 17.2).

Example 15.4 For a sodium lamp, the distance traversed by the mirror between two successive disappearances is 0.289 mm. Calculate the difference in the wavelengths of the D_1 and the D_2 lines. Assume λ = 5890 Å.

Solution: When the mirror moves through a distance 0.289 mm, the additional path introduced is 0.578 mm. Thus,

$$\frac{0.578}{\lambda} - \frac{0.578}{\lambda + \Delta\lambda} = 1$$

or

$$\Delta\lambda \simeq \frac{\lambda^2}{0.578} = \frac{(5890 \times 10^{-7})^2}{0.578} \text{ mm}$$
$$\simeq 6 \text{ Å}$$

Summary

- If a plane wave is incident normally on a thin film of uniform thickness d then the waves reflected from the upper surface interfere with the waves reflected from the lower surface. Indeed, for a film of thickness $\lambda/4n_f$ [where λ is the free space wavelength and n_f is the film refractive index which lies between the refractive indices of the two surrounding media], the wave reflected from the upper surface interferes destruc-

* The zero path difference is easily obtained by using white light where only a few coloured fringes, around $d = 0$, will be visible.